Chemistry 3A

Introductory General Chemistry

Experiment 0a

Lab Exploration

Mitch Halloran

Fresno City College

Special Guidance Today

- Because you did not have access to your laboratory manuals for this start of lab session, I will quickly guide you to what you need to pay attention to
- You will have your manuals today for reference
- You have a data report form you fill out today
- Use ink pen. NO PENCILS. Just cross-out mistakes

- There are 3 parts to Expt 0a
- We skip Part 1 – Scavenger Hunt
- It has you looking for 20 items; we can learn this later

Part 2: Using A Buret

Equipment/Material Needed	
Buret *handle carefully. It is an easily breakable analytical device!*	
Buret clamp	50 mL beaker
50 mL graduated cylinder	Erlenmeyer flask

Consumables Needed	
DI water (~60 mL)	food coloring (2 dyes)

- ❖ Page 12: mix 1 gtt food dye in ~30 mL DI H_2O in beaker; fill buret; record initial reading on buret; record vol + sketch
- ❖ p 13: fill grad cylinder ~20-30 mL DI H_2O; record vol + sketch; pour in Erlenmeyer; add different food dye color; mix
- ❖ from buret, add ~ 5 mL dyed H_2O to Erlenmeyer flask while swirling to mix (DON'T LET FLASK HIT BURET TIP)
- ❖ Buret stopcock control skill: 1 mL volume + droplet formation

Part 3: Using Pipets, Volumetric Flasks

Equipment/Material Needed

50 mL volumetric flask (2) w/ stopper	50 mL beaker
scoopula	10.00 mL volumetric pipet
balance	

Consumables Needed

4-6 g NaCl (sodium chloride)	food coloring (2 dyes)
DI water	

- Page 14: tare balance; weigh empty 50 mL stoppered vol. flask; record to data report
- Tare 50 mL beaker; add NaCl; record mass
- Add ~10 mL DI water; swirl to dissolve as much as possible; transfer to 50 mL vol flask; add gtt dye; add more H_2O, follow transfer method; fill to flask mark w/ transfer pipet
- Take 10.00 mL from flask to new 50 ml flask (use <u>bulb</u> carefully). Fill to flask mark as before using transfer pipet
- Compare color of solutions, fill out data report

Example Data Analysis

Exp 0a (Intro) *Lab Exploration*

DATA AND REPORT – DUE ____ / ____ / ____

SECTION ________________

DATE ________________

PARTNER ________________

Did you put your partner's name?

A significant cause of point loss is failing to write in the partner's name. Don't let it happen to you

PART 1: SCAVENGER HUNT

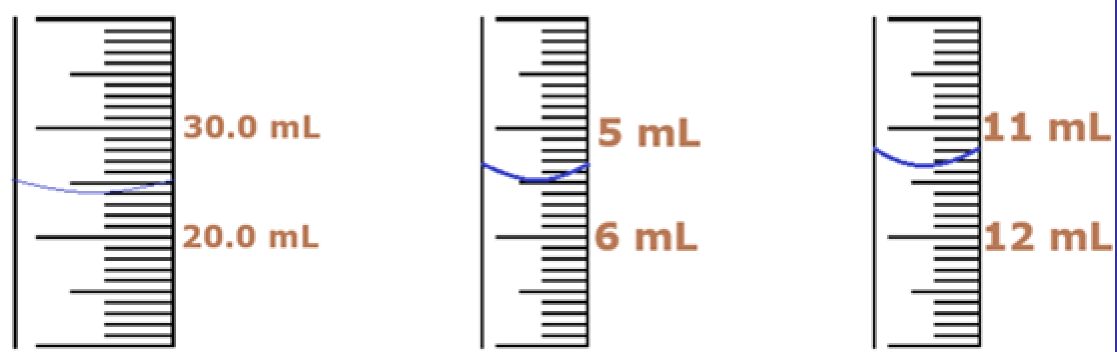
Instructor Check: ______________

DATA TABLE – PART 2: USING A BURET

Measurements	
Graduated Cylinder	24.0 mL
Initial Buret Reading:	5.50 mL
Final Buret Reading:	11.35 mL
Calculations	
Volume Delivered by Buret	11.35 – 5.50 = 5.85 mL
Lab-Partner Check	
Adding Dropwise	✓
Adding Partial Drop	✓

SKETCH OF MENISCUS LOCATION

Graduated Cylinder | Initial Buret Reading: | Final Buret Reading:



Example Data Analysis

Lab Safety Contract

DATA TABLE – PART 3: USING VOLUMETRIC FLASKS AND PIPETS

Measurements	
Mass of 50-mL Volumetric Flask + stopper (empty)	72.3049 g
Mass of sodium chloride (NaCl, salt)	4.7015 g
Mass of 50-mL Volumetric Flask + stopper + stock solution	125.1611 g
Visual comparison of stock solution and diluted solution	Observations: diluted solution much lighter in color (20% as intense as "stock solution")
Calculations	
Mass of stock solution made *(salt and water)*	125.1611 – 72.3049 = 52.8562 g solution
Mass of water used to make the stock solution	52.8562 – 4.7015 = 48.1547 g water
Lab-Partner Check	
Pipet technique	✓

Clean Up

- Water and salt solutions can safely be poured down the drain. (Like any household solutions)

  This will not be the case for many solutions which may be strong acids & bases or certain inorganic compounds/salts

- Glassware cleaning:
  - First with a mild detergent solution: a precaution for organic substances that are lipophilic
  - Final rinsing with DI water a couple of times:
    - 1st rinse with a very small volume
    - 2nd rinse with a somewhat bigger volume
- The best drying of water-rinsed containers and equipment is air drying. You can usually properly place inverted glassware and other equipment on the angled peg drying racks.